

Submission Format

Submit **one CSV file** with exactly **88 lines** (and the github repo link; make it public :)): a header row plus one row for each of the 87 stars in `private_pack.zip` .

Columns

Column	Type	Required	Description
<code>star_id</code>	string	always	Filename without extension, e.g. <code>STAR_0043</code>
<code>prediction</code>	0 or 1	always	1 if you believe a transiting planet is present
<code>confidence</code>	float 0–1	always	Your belief that a planet is present. Used for ranking.
<code>period</code>	float, days	when <code>prediction = 1</code>	Orbital period of the detected signal
<code>depth_ppm</code>	float, ppm	when <code>prediction = 1</code>	Transit depth in parts per million
<code>duration_hours</code>	float, hours	when <code>prediction = 1</code>	Transit duration

Example

```
star_id,prediction,confidence,period,depth_ppm,duration_hours
STAR_0000,1,0.94,35.4184,1976.0,6.18
STAR_0001,0,0.03,,,
STAR_0002,1,0.61,12.3341,285.5,3.92
STAR_0003,0,0.11,,,
```

Leave the last three fields **empty** when `prediction = 0` . Do not write `NA` , `null` , `-1` , or `0` .

Rules

- Exactly one row per star. All 87 `star_id` values must appear, none twice.
- `star_id` must match the filename exactly, including the `STAR_` prefix and four digits. `STAR_0007` is valid; `star_0007` , `STAR_7` , `7` are not.
- `confidence` must be a real number between 0 and 1. **Do not set every row to 1.0** — ranking metrics are a large part of the score, and flat confidence values score poorly.
- `confidence` should be consistent with `prediction` , but they are scored separately. A star with `prediction = 0` and `confidence = 0.45` is legal and means “probably nothing, but not certain”.
- UTF-8, comma-separated, no index column, no trailing blank lines.

How your submission is scored

Detection — computed over all 87 stars.

- Precision, recall and F1 from the `prediction` column
- PR-AUC and average precision from the `confidence` column

The `confidence` ranking matters as much as the hard `prediction` . A pipeline that is confident about obvious transits and uncertain about marginal ones will outrank one that guesses the same number of stars with flat confidence.

Characterisation — computed over the subset of stars where a known transit signal exists and you predicted 1.

- Period accuracy: correct if within 2% of the true period, or of a $2\times$, $\frac{1}{2}\times$, $3\times$ or $\times$ alias of it
- Depth accuracy: relative error against the true depth

Reporting a period you did not actually measure will not help. Random values almost never fall inside the tolerance, and a wrong period costs you nothing beyond the credit you would have earned for a right one.

Difficulty breakdown — the private set spans a range of transit depths, from easily visible to near the noise floor. Scores are reported per difficulty band as well as overall, so recovering shallow signals is worth substantially more than recovering obvious ones.

Validate before you submit

Run this and fix anything it prints. A malformed file may be rejected outright.

```
import pandas as pd

sub = pd.read_csv("submission.csv")
need = ["star_id", "prediction", "confidence",
        "period", "depth_ppm", "duration_hours"]

assert list(sub.columns) == need, f"columns must be exactly {need}"
assert len(sub) == 87, f"expected 87 rows, got {len(sub)}"
assert sub.star_id.nunique() == 87, "duplicate star_id"
assert sub.star_id.str.match(r"^STAR_\d{4}$").all(), "bad star_id format"
assert sub.prediction.isin([0, 1]).all(), "prediction must be 0 or 1"
assert sub.confidence.between(0, 1).all(), "confidence must be in [0, 1]"

pos = sub[sub.prediction == 1]
for c in ("period", "depth_ppm", "duration_hours"):
    assert pos[c].notna().all(), f"{c} missing for some prediction=1 rows"
assert (pos.period > 0).all(), "period must be positive"

print(f"OK — {len(pos)} detections, {len(sub) - len(pos)} non-detections")
print(f"confidence: min {sub.confidence.min():.3f}, "
      f"max {sub.confidence.max():.3f}, "
      f"unique {sub.confidence.nunique()}")
```

If `unique` comes back as 1 or 2, your confidence column is not carrying information and you are giving away a large part of the score.

Filename

`submission_<teamname>.csv` — lowercase team name, no spaces.